Lecture 19

Lorentzian geometry and general relativity

Objectives. *A Lorentzian metric specifies elapsed time and the paths of freely falling particles. Einstein's equation relates its curvature to matter. We recover Newton's acceleration in a weak field and verify the Schwarzschild vacuum metric. Its conserved quantities then reduce a massive particle's motion to an orbit equation.*

By the end of this lecture you will be able to:

- Distinguish timelike, null and spacelike tangent vectors in signature $+---$.
- Recover Newton's equation from a weak static metric and slow motion.
- Explain the action and conservation statements in Einstein's equation.
- Verify Schwarzschild vacuum curvature and derive the massive orbit equation.

Unit 19A Motion in a given metric

19.1 Lorentzian metrics and proper time

Definition 19.1 (Lorentzian spacetime)**.** A *Lorentzian spacetime* is a smooth four-dimensional manifold with a metric of signature $(+,-,-,-)$.

At each point an orthonormal frame identifies the metric with $\eta = \mathrm{diag}(1,-1,-1,-1)$. This recalls the flat interval in *Symmetries in Physics*, Lecture 17, *Minkowski space from physics*. Here the metric may vary from point to point. Assume the spacetime is oriented, so its metric volume form is defined.

Definition 19.2 (Causal type and time orientation)**.** A nonzero tangent vector v is *timelike* if $g(v,v) > 0$, *null* if $g(v,v) = 0$, and *spacelike* if $g(v,v) < 0$. The null vectors form the *light cone*. A *time orientation* is a continuous choice of one of the two timelike cones as future.

We assume such a choice. In an orthonormal frame adapted to this choice, future vectors have positive time component. The figure shows one space direction and the time direction.

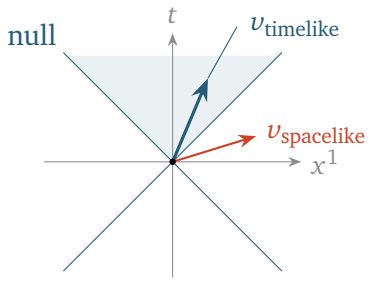


Figure 19.1: The null lines bound the future timelike cone in a tangent plane. The classification uses the metric at the point, rather than the drawing's Euclidean angles.

We use $c = 1$ and coordinates $x^0 = t$.

Definition 19.3 (Proper time and four-velocity)**.** The *proper time* along a future timelike curve $x(\sigma)$ is defined by

$$\tau(\sigma_1) - \tau(\sigma_0) = \int_{\sigma_0}^{\sigma_1} \sqrt{g_{\mu\nu} \frac{\mathrm{d}x^\mu}{\mathrm{d}\sigma} \frac{\mathrm{d}x^\nu}{\mathrm{d}\sigma}}\, \mathrm{d}\sigma.$$

Its *four-velocity* is $U = \mathrm{d}x/\mathrm{d}\tau$.

The definition gives $g(U, U) = 1$.

Physical input. The physical free-fall postulate says that an unforced test particle follows a timelike geodesic of the Levi-Civita connection. A light ray follows a null geodesic, using an affine parameter in place of proper time. These laws describe unforced structureless test bodies and light in the geometric-optics approximation. Universal free fall motivates assigning test-body trajectories to spacetime geometry. It does not derive the metric geodesic law. An ideal clock measures proper time along its worldline.

The existence and equation of these geodesics are the mathematical results of Lecture 16.

19.2 Newton's equation from a weak metric

Let $\Phi = \epsilon \varphi_{\text{pot}}(x^1, x^2, x^3)$ be a smooth static potential, where ϵ is a small parameter. Derivatives of Φ count as first order in ϵ. We write the metric as $\mathrm{d}s^2 = g_{\mu\nu}\mathrm{d}x^\mu \mathrm{d}x^\nu$. Products denote symmetric tensor products, so $\mathrm{d}t^2 = \mathrm{d}t \otimes \mathrm{d}t$ and $2\mathrm{d}v\,\mathrm{d}r = \mathrm{d}v \otimes \mathrm{d}r + \mathrm{d}r \otimes \mathrm{d}v$. In spatial Cartesian coordinates consider

$$\mathrm{d}s^2 = (1 + 2\Phi)\mathrm{d}t^2 - (1 - 2\Phi) \sum_{i=1}^{3} (\mathrm{d}x^i)^2.$$

Assume $|\Phi| < 1/2$, so the signature is Lorentzian. A clock at fixed spatial coordinates has rate

$$\frac{\mathrm{d}\tau}{\mathrm{d}t} = \sqrt{1 + 2\Phi} = 1 + \Phi + O(\epsilon^2).$$

Thus a clock at a lower potential accumulates less proper time over the same coordinate-time interval. The time coefficient determines the leading slow-motion acceleration. The spatial factor also enters curvature and the field equation in §19.3. We define the Euclidean spatial gradient by

$$\nabla_E \Phi = (\partial_1 \Phi, \partial_2 \Phi, \partial_3 \Phi).$$

The figure shows a trajectory in its leading Newtonian approximation.

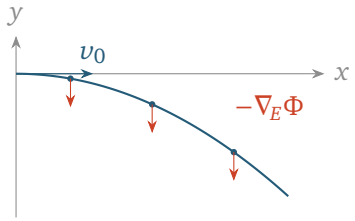


Figure 19.2: For a locally uniform downward acceleration, the leading Newtonian trajectory is $x = v_0 t$, $y = -at^2/2$, with $v_0 > 0$ and $a = |\nabla_E \Phi| > 0$. This curve represents the weak-field, slow-motion limit.

Proposition 19.4 (Weak-field free fall · *proved*). *For a future timelike geodesic with small coordinate speed, the coordinate-time acceleration to first order in the potential and lowest order in speed is*

$$\frac{\mathrm{d}^2 x^i}{\mathrm{d}t^2} = -\partial_i \Phi, \qquad i = 1, 2, 3.$$

Proof. We use an affine parameter λ and write dots for its derivatives. Put $v^\mu = \mathrm{d}x^\mu/\mathrm{d}t$, so $v^0 = 1$. A timelike tangent has nonzero time component in this metric, since a nonzero tangent with zero time component has negative metric square. The chain rule and the affine time equation give

$$\ddot{x}^i = \frac{\mathrm{d}^2 x^i}{\mathrm{d}t^2}\dot{t}^2 + v^i \ddot{t}, \qquad \ddot{t} = -\Gamma^0_{\mu\nu} v^\mu v^\nu \dot{t}^2.$$

We substitute these identities in the spatial geodesic equation and divide by $\dot{t}^2$. This gives

$$\frac{\mathrm{d}^2 x^i}{\mathrm{d}t^2} = -\Gamma^i_{\mu\nu} v^\mu v^\nu + v^i \Gamma^0_{\mu\nu} v^\mu v^\nu.$$

The Christoffel formula gives

$$\Gamma^i_{00} = \frac{\partial_i \Phi}{1 - 2\Phi} = \partial_i \Phi + O(\epsilon^2), \qquad \Gamma^0_{0i} = \frac{\partial_i \Phi}{1 + 2\Phi}.$$

The static diagonal metric has $\Gamma^0_{00} = \Gamma^0_{ij} = \Gamma^i_{0j} = 0$. Write $\mathbf{v} = (v^1, v^2, v^3)$ for the coordinate spatial velocity. All remaining terms in the spatial equation contain at least two spatial velocity factors. The acceleration is $-\partial_i \Phi + O(\epsilon^2) + O(\epsilon |\mathbf{v}|^2)$, which proves the stated approximation. ∎

Physical input. For an attractive Newtonian point source of mass $m > 0$, $\Phi = -G_N m/r$, where $G_N > 0$ is Newton's constant. For static mass density ρ, Newton's field law is $\nabla_E^2 \Phi = 4\pi G_N \rho$.

Its negative gradient points toward the source.

Two nearby freely falling particles reveal the second derivatives of the potential. Let ξ^i be their infinitesimal separation in the weak static field. Subtracting their leading Newtonian equations and linearizing in separation gives

$$\ddot{\xi}^i = -\partial_i \partial_j \Phi \, \xi^j.$$

Here dots denote coordinate-time derivatives. Lecture 17's curvature formula gives, to first order in ϵ,

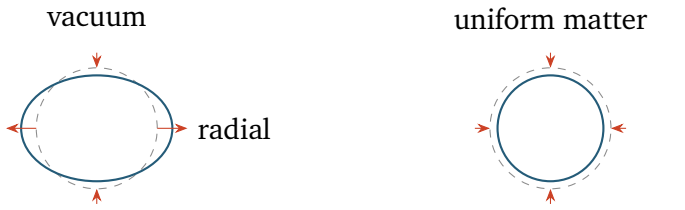
$$R^i{}_{0j0} = \partial_j \Gamma^i_{00} = \partial_i \partial_j \Phi, \qquad \ddot{\xi}^i = -R^i{}_{0j0} \xi^j.$$

These are tidal relative accelerations in the static weak-field, slow-motion approximation. Taking the trace gives $\mathrm{Ric}_{00} = \nabla_E^2\Phi$ at this order. For an initially comoving small ball of particles with volume V, the separation map at short elapsed time Δt is $\mathbf{1} - \frac{1}{2}(\Delta t)^2\,\mathrm{Hess}_E\,\Phi$ to second order. Differentiating its determinant at $\Delta t = 0$ gives

$$\left.\frac{\dot V}{V}\right|_{\Delta t=0} = 0, \qquad \left.\frac{\ddot V}{V}\right|_{\Delta t=0} = -\nabla_E^2\Phi.$$

The ball can deform even when this trace vanishes.

For the exterior potential $-G_N m/r$, the Hessian eigenvalues are $-2G_N m/r^3$ radially and $G_N m/r^3$ in each transverse direction. The relative acceleration therefore stretches the ball radially and squeezes it transversely. For comparison, a uniform-density Newtonian model has local potential $\Phi = 2\pi G_N \rho r^2/3 + \text{constant}$. Its Hessian is $(4\pi G_N\rho/3)\mathbf{1}$. For the drawing, we evaluate the vacuum map at $\Delta t\sqrt{G_N m/r^3} = 1/2$ and the matter map at $\Delta t\sqrt{4\pi G_N\rho/3} = 1/2$. The respective axis factors are $(5/4, 7/8, 7/8)$ and $(7/8, 7/8, 7/8)$. The figure compares these second-order separation maps.



$\mathrm{Hess}_E\,\Phi = \frac{G_N m}{r^3}\,\mathrm{diag}(-2,1,1)$, $\mathrm{tr}(\mathrm{Hess}_E\,\Phi) = 0$

$\mathrm{Hess}_E\,\Phi = \frac{4\pi G_N\rho}{3}\mathbf{1}$, $\mathrm{tr}(\mathrm{Hess}_E\,\Phi) = 4\pi G_N\rho$

Figure 19.3: Dashed circles show initially comoving balls, solid curves their second-order separation maps, and arrows the relative acceleration in the weak static slow-motion approximation. The Hessian trace is zero in vacuum and $4\pi G_N\rho$ in uniform matter.

Unit 19B Einstein's equation and matter

19.3 Einstein's equation and its conservation identity

The given weak metric predicts motion. A field law must also constrain the metric from the energy and momentum of matter.

Definition 19.5 (Stress–energy tensor)**.** The symmetric *stress–energy tensor* T describes energy density, momentum density and their fluxes. In an orthonormal observer frame with $c = 1$, its upper components have the meanings

T^{00}	energy density
$T^{0i} = T^{i0}$	energy flux in direction i and momentum density
T^{ij}	flux of i-momentum across a j-normal face.

Stress includes pressure, the transport of normal momentum across a face. Lowering with signature $+---$ gives $T_{00} = T^{00}$, $T_{0i} = -T^{0i}$ and $T_{ij} = T^{ij}$. The figure fixes the face normal used to read a flux component.

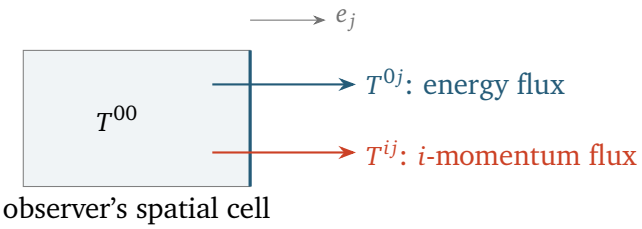


Figure 19.4: An observer's cell stores energy density T^{00}. Across its face of outward normal e_j, energy flux is T^{0j} and the flux of i-momentum is T^{ij}, in an orthonormal frame with $c = 1$.

Physical input. For pressureless matter (dust), the model is

$$T_{\mu\nu} = \rho U_\mu U_\nu, \qquad g(U,U) = 1.$$

Here ρ is rest energy density and U is the matter four-velocity. In its rest frame the stress tensor is $\operatorname{diag}(\rho, 0, 0, 0)$.

Recall Lecture 17's definitions $\mathrm{Ric}_{\mu\nu} = R^\rho{}_{\mu\rho\nu}$ and $R_{\mathrm{sc}} = g^{\mu\nu}\,\mathrm{Ric}_{\mu\nu}$.

Definition 19.6 (Einstein tensor)**.** The *Einstein tensor* is

$$G_{\mu\nu} = \mathrm{Ric}_{\mu\nu} - \frac{1}{2} R_{\mathrm{sc}} g_{\mu\nu}.$$

Physical input. We use Einstein's field law in four dimensions with zero cosmological constant:

$$G_{\mu\nu} = 8\pi G_N T_{\mu\nu}.$$

Conservation and the Newtonian limit motivate and check this choice. They do not prove that it is the unique possible gravitational field law.

The conservation identity follows from the covariant second Bianchi identity of Proposition 17.25, whose conversion from forms was sketched. We write $\nabla_\lambda T_{\dots} = (\nabla_{\partial_\lambda} T)_{\dots}$ and $\nabla^\mu = g^{\mu\lambda}\nabla_\lambda$. We contract the identity as follows. Use the lowered tensor $R_{ab\mu\nu}$ of Lecture 17. Metric compatibility and the two pair antisymmetries give

$$\begin{aligned} 0 &= g^{a\mu} g^{b\lambda} \left(\nabla_\lambda R_{ab\mu\nu} + \nabla_\mu R_{ab\nu\lambda} + \nabla_\nu R_{ab\lambda\mu} \right) \\ &= 2\nabla^\mu \mathrm{Ric}_{\mu\nu} - \partial_\nu R_{\mathrm{sc}}. \end{aligned}$$

The three contractions are respectively $\nabla^\mu \mathrm{Ric}_{\mu\nu}$, $\nabla^\mu \mathrm{Ric}_{\mu\nu}$ and $-\partial_\nu R_{\mathrm{sc}}$. Thus $\nabla^\mu G_{\mu\nu} = 0$. Einstein's equation therefore implies $\nabla^\mu T_{\mu\nu} = 0$. This is local covariant conservation. It alone does not assert a globally conserved energy on an arbitrary spacetime. For a Killing field K, the current $J_\nu = T_{\mu\nu} K^\mu$ obeys

$$\nabla^\nu J_\nu = T_{\mu\nu} \nabla^\nu K^\mu = 0.$$

The last equality contracts a symmetric tensor with the antisymmetric Killing derivative from Lecture 16. This supplies a conserved flux through hypersurfaces when the boundary flux vanishes and the integrals exist.

For dust, the product rule expands conservation as

$$0 = \nabla^\mu (\rho U_\mu U_\nu) = U_\nu \nabla_\mu (\rho U^\mu) + \rho U^\mu \nabla_\mu U_\nu.$$

Contracting with U^ν removes the acceleration term because metric compatibility gives $U^\nu U^\mu \nabla_\mu U_\nu = \frac{1}{2} U^\mu \nabla_\mu g(U,U) = 0$. Thus $\nabla_\mu(\rho U^\mu) = 0$. Where $\rho > 0$, substitution back gives $\nabla_U U = 0$. The dust conservation equation therefore yields both rest-density continuity and geodesic matter motion.

For the weak metric above, the additional first-order symbols are

$$\Gamma^i_{jk} = -\delta^i_k \partial_j \Phi - \delta^i_j \partial_k \Phi + \delta_{jk} \partial_i \Phi, \qquad \Gamma^\mu_{\mu i} = -2\partial_i \Phi.$$

Here and in the next calculation terms of order ϵ^2 are omitted. The quadratic Christoffel terms do not contribute at first order. Staticity removes time derivatives and gives $\mathrm{Ric}_{0i} = 0$. Thus $\mathrm{Ric}_{ij} = \partial_k \Gamma^k_{ji} - \partial_j \Gamma^\mu_{\mu i}$: the mixed second derivatives cancel, leaving $\delta_{ij} \sum_k \partial_k^2 \Phi$. The full contraction gives

$$\mathrm{Ric}_{00} = \nabla_E^2 \Phi, \qquad \mathrm{Ric}_{ij} = \delta_{ij} \nabla_E^2 \Phi, \qquad R_{\mathrm{sc}} = -2\nabla_E^2 \Phi, \qquad G_{00} = 2\nabla_E^2 \Phi.$$

Here $\nabla_E^2 = \sum_i \partial_i^2$ is the Euclidean spatial Laplacian. For example, at this order $\mathrm{Ric}_{00} = \partial_i \Gamma^i_{00}$. The 00 field equation for slowly moving matter gives $\nabla_E^2 \Phi = 4\pi G_N \rho$, the positive Newtonian Poisson equation. This checks both the sign and the coefficient of the field law.

The spatial perturbation in the weak metric contributes to this check. If we retained $g_{00} = 1 + 2\Phi$ but set $g_{ij} = -\delta_{ij}$, the linearized Einstein component G_{00} would instead vanish. The leading slow-motion acceleration alone does not determine the spatial metric.

19.3.1 The variational form

We take action integrals on a relatively compact coordinate domain. All variations vanish near its boundary. Let S_{m} be a matter action and let $\mu = \sqrt{|\det g|}\, \mathrm{d}t \wedge \mathrm{d}x^1 \wedge \mathrm{d}x^2 \wedge \mathrm{d}x^3$ in positive coordinates.

Physical input. We normalize the matter action so that its response to compactly supported inverse-metric variations, with matter fixed, is

$$\delta S_{\mathrm{m}} = \frac{1}{2} \int T_{\alpha\beta}\, \delta g^{\alpha\beta} \mu.$$

This identifies the variational tensor with the physical stress–energy tensor.

Write $F_{\mu\nu}$ for the components of F_{em} in this paragraph.

Physical input. For the field of Lecture 15, we use the normalized Maxwell action

$$S_{\mathrm{em}} = -\frac{1}{2} \int F_{\mathrm{em}} \wedge \star F_{\mathrm{em}} = -\frac{1}{4} \int F_{\rho\sigma} F^{\rho\sigma} \mu.$$

This is Nakahara's (7.210b) in the present units.

Vary the inverse metric while fixing the potential, hence the covariant field components. The inverse-metric factors and the determinant variation give

$$\delta(F_{\rho\sigma} F^{\rho\sigma}) = 2 F_{\alpha\lambda} F_\beta{}^\lambda \delta g^{\alpha\beta}, \qquad \delta\mu = -\frac{1}{2} g_{\alpha\beta} \delta g^{\alpha\beta} \mu.$$

Substitution in the action and comparison with the stress definition give

$$T_{\mu\nu} = -F_{\mu\lambda} F_\nu{}^\lambda + \frac{1}{4} g_{\mu\nu} F_{\alpha\beta} F^{\alpha\beta}.$$

In an orthonormal frame, $F_{0\lambda}F_0{}^{\lambda} = -|\mathbf{E}|^2$ and $F_{\alpha\beta}F^{\alpha\beta} = 2(|\mathbf{B}|^2 - |\mathbf{E}|^2)$. Hence $T_{00} = (|\mathbf{E}|^2 + |\mathbf{B}|^2)/2$ is nonnegative.

Physical input. We use the Einstein–Hilbert gravitational action

$$S_{\text{grav}} = -\frac{1}{16\pi G_N}\int R_{\text{sc}}\mu.$$

Theorem 19.7 (Variational form of Einstein's equation · *cited variation; algebraic consequence proved*)**.** *For compactly supported metric variations, the cited variation formula is*

$$\delta\int R_{\text{sc}}\mu = \int G_{\alpha\beta}\delta g^{\alpha\beta}\mu.$$

Stationarity of $S_{\text{grav}} + S_{\text{m}}$ *therefore gives*

$$\text{Ric}_{\mu\nu} - \frac{1}{2}R_{\text{sc}}g_{\mu\nu} = 8\pi G_N T_{\mu\nu}.$$

The variation formula is Nakahara, §7.10.2, equations (7.203)–(7.214), translated to our signature and matter convention.[1]

It discards a divergence because the variation has compact support. The displayed coefficient follows by setting $-G_{\mu\nu}/(16\pi G_N) + T_{\mu\nu}/2 = 0$. The action gives the stated physical law its variational form.

Self-check. State in one sentence what the Einstein tensor is and why we introduced it.

Answer. The Einstein tensor is the trace-adjusted Ricci tensor $G_{\mu\nu} = \text{Ric}_{\mu\nu} - \frac{1}{2}R_{\text{sc}}g_{\mu\nu}$, introduced as a curvature source tensor whose divergence vanishes. The decisive identity is $2\nabla^{\mu}\,\text{Ric}_{\mu\nu} = \partial_\nu R_{\text{sc}}$, which cancels the trace term's divergence by metric compatibility.

Unit 19C Schwarzschild geometry and orbits

19.4 The Schwarzschild vacuum metric

In vacuum, $T_{\mu\nu} = 0$ makes the Einstein tensor zero. Taking the metric trace in four dimensions gives $0 = R_{\text{sc}} - \frac{1}{2}(4)R_{\text{sc}} = -R_{\text{sc}}$. Thus the vacuum field equation is $\text{Ric}_{\mu\nu} = 0$.

We now use geometric units $G_N = c = 1$. Let $M > 0$ denote the mass parameter, equal to $G_N m/c^2$ before this unit choice. On the exterior $r > 2M$, set $f(r) = 1 - 2M/r$.

Definition 19.8 (Schwarzschild metric)**.** The *Schwarzschild metric* is

$$\mathrm{d}s^2 = f\,\mathrm{d}t^2 - f^{-1}\mathrm{d}r^2 - r^2(\mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\phi^2).$$

[1] Replacing a metric by its negative leaves its Christoffels, curvature operator and Ricci tensor unchanged. The scalar curvature reverses sign, so the Einstein tensor is unchanged. Nakahara's − + ++ Einstein–Hilbert sign consequently becomes the minus sign above. For stress, his (7.211) varies the lower metric. Converting to his inverse metric contributes a minus sign by (7.203). Our inverse metric is its negative, giving a second minus sign. The two cancel, yielding our positive inverse-metric variation with the same lowered stress tensor. We use (7.211) for this translation. The inverse-metric prefactor in (7.212) needs a minus sign to agree with it. Carroll's free notes use the same curvature index order as ours. Carroll, *Lecture Notes on General Relativity* (1997), (3.67), (3.90)–(3.96), pp. 76, 81–82. His metric has the opposite sign. His Schwarzschild geodesic formulas (7.33)–(7.48), pp. 172–174, translate by this metric reversal.

The angular expression is the round-sphere metric, with usual overlapping charts at its coordinate poles.

Proposition 19.9 (Schwarzschild vacuum). *The exterior Schwarzschild metric has* $\mathrm{Ric} = 0$. *The fields* ∂_t *and* ∂_ϕ *are Killing fields. Along an affinely parametrized geodesic, the quantities*

$$E = f\dot{t}, \qquad L = r^2 \sin^2\theta\, \dot{\phi}$$

are constant.

Proof. First compute for any smooth positive function $f(r)$. Up to symmetry in the two lower indices, the nonzero Christoffels are

$$\begin{aligned} &\Gamma^t_{tr} = f'/(2f), && \Gamma^r_{tt} = ff'/2, && \Gamma^r_{rr} = -f'/(2f),\\ &\Gamma^r_{\theta\theta} = -rf, && \Gamma^r_{\phi\phi} = -rf\sin^2\theta, && \Gamma^\theta_{r\theta} = 1/r,\\ &\Gamma^\phi_{r\phi} = 1/r, && \Gamma^\theta_{\phi\phi} = -\sin\theta\cos\theta, && \Gamma^\phi_{\theta\phi} = \cot\theta. \end{aligned}$$

Substitute them into Lecture 17's Ricci contraction,

$$\mathrm{Ric}_{\mu\nu} = \partial_\rho\Gamma^\rho_{\nu\mu} - \partial_\nu\Gamma^\rho_{\rho\mu} + \Gamma^\rho_{\rho\lambda}\Gamma^\lambda_{\nu\mu} - \Gamma^\rho_{\nu\lambda}\Gamma^\lambda_{\rho\mu}.$$

For example, $\Gamma^\rho_{\rho r} = 2/r$ and the tt entry is

$$\mathrm{Ric}_{tt} = \frac{(f')^2 + ff''}{2} + \frac{ff'}{r} - \frac{(f')^2}{2} = f\left(\frac{f''}{2} + \frac{f'}{r}\right).$$

For the other independent diagonal entries, substitution gives

$$\begin{aligned} \mathrm{Ric}_{rr} &= \partial_r\left(-\frac{f'}{2f}\right) + \frac{2}{r^2} - \frac{f'}{rf} - \left(\frac{(f')^2}{2f^2} + \frac{2}{r^2}\right),\\ \mathrm{Ric}_{\theta\theta} &= \partial_r(-rf) + \csc^2\theta - 2f - (-2f + \cot^2\theta). \end{aligned}$$

The off-diagonal entries vanish. Simplifying the diagonal entries gives

$$\mathrm{Ric}_{rr} = -f^{-1}\left(\frac{f''}{2} + \frac{f'}{r}\right), \qquad \mathrm{Ric}_{\theta\theta} = 1 - f - rf', \qquad \mathrm{Ric}_{\phi\phi} = \sin^2\theta\ \mathrm{Ric}_{\theta\theta}\,.$$

The equation $\mathrm{Ric}_{\theta\theta} = 0$ is $(rf)' = 1$. Integration gives $f = 1 - C/r$ for a constant C. This also satisfies $f''/2 + f'/r = 0$. Within this one-function ansatz, the vacuum equations therefore determine f. Comparison with $g_{00} = 1 + 2\Phi$ and $\Phi = -G_N m/r$ in units $c = 1$ identifies $C = 2G_N m = 2M$. This explains the mass parameter. It is not a proof that every static spherically symmetric metric has the assumed form. For $f = 1 - 2M/r$, the combinations are $f''/2 + f'/r = -2M/r^3 + 2M/r^3 = 0$ and $1 - f - rf' = 2M/r - 2M/r = 0$. Thus every Ricci component is zero. Smoothness extends this tensor equality across the angular chart poles.

The metric coefficients are independent of t and ϕ. The Lie derivative identity in Lecture 16, §16.4, is

$$(\mathcal{L}_K g)(Y,Z) = Kg(Y,Z) - g([K,Y],Z) - g(Y,[K,Z]).$$

For $K = \partial_t$ or $K = \partial_\phi$ and coordinate fields Y and Z, the brackets vanish. Independence of the metric coefficients therefore gives $\mathcal{L}_{\partial_t} g = \mathcal{L}_{\partial_\phi} g = 0$. Lecture 16's Killing conservation theorem makes $g(\partial_t, \dot{\gamma}) = f\dot{t}$ and $-g(\partial_\phi, \dot{\gamma}) = r^2\sin^2\theta\dot{\phi}$ constant. ∎

The vacuum exterior is not flat. Lecture 17's identity Ric = Kg applied only to surfaces. Here the Christoffel table gives

$$\begin{aligned} R^{\theta}{}_{\phi\theta\phi} &= (\sin^2\theta - \cos^2\theta) - f\sin^2\theta + \cos^2\theta \\ &= (1-f)\sin^2\theta = \frac{2M}{r}\sin^2\theta. \end{aligned}$$

This component is nonzero for $M > 0$ and $0 < \theta < \pi$. The Ricci contraction vanishes while the full curvature remains nonzero.

The divergence of f^{-1} at $r = 2M$ is a coordinate singularity. Indeed, put $v = t + r_*(r)$, where $r_*' = f^{-1}$. Substitution of $\mathrm{d}t = \mathrm{d}v - f^{-1}\mathrm{d}r$ gives

$$\mathrm{d}s^2 = f\,\mathrm{d}v^2 - 2\mathrm{d}v\,\mathrm{d}r - r^2(\mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\phi^2).$$

Its determinant is $-r^4\sin^2\theta$, nonzero at $r = 2M$ away from the angular coordinate poles. These coordinates extend the metric there. Radial null directions satisfy

$$0 = \mathrm{d}v(f\,\mathrm{d}v - 2\mathrm{d}r).$$

The ingoing family has $\mathrm{d}v = 0$, with future direction $\mathrm{d}r < 0$. The outgoing family has $\mathrm{d}r/\mathrm{d}v = f/2$, with $\mathrm{d}v > 0$. The future outgoing slope is positive outside $r = 2M$, zero on it and negative inside. Both future null directions decrease r inside. Their future radial cone contains no direction of increasing r. Angular motion contributes a nonpositive term to the metric, so a future causal direction with $\mathrm{d}v > 0$ satisfies $\mathrm{d}r/\mathrm{d}v \le f/2$. It also decreases r inside. The figure shows these local null directions in the extended coordinates.

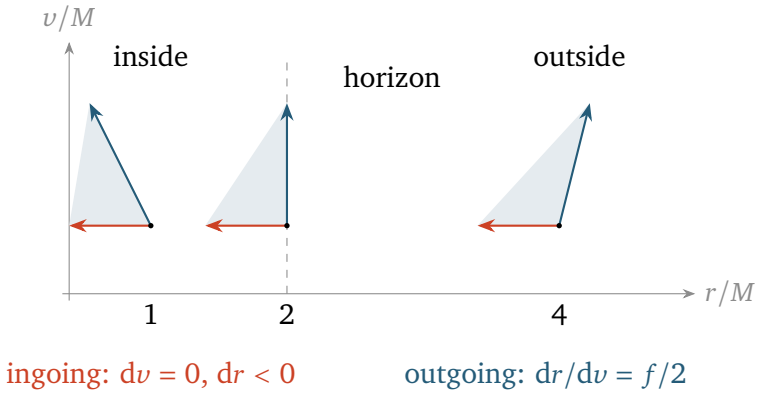


Figure 19.5: With v vertical and r horizontal, the future ingoing ray points left and the future outgoing ray has slope $\mathrm{d}r/\mathrm{d}v = f/2$. The shaded local future cones tilt inward for $r < 2M$.

Definition 19.10 (Event horizon)**.** The future *event horizon* is the boundary of the region from which future-directed light signals can reach arbitrarily distant observers.

In the Schwarzschild black-hole extension, the surface $r = 2M$ is this horizon. This global identification is cited from Carroll, §7, pp. 183–185. The local cone calculation establishes its one-way property. The curvature singularity at $r = 0$ can be detected by a scalar. In the exterior $r > 2M$, use the orthonormal frame

$$f^{-1/2}\partial_t, \qquad f^{1/2}\partial_r, \qquad r^{-1}\partial_\theta, \qquad (r\sin\theta)^{-1}\partial_\phi.$$

Label these vectors by 0, 1, 2 and 3 in the displayed order. Substituting the Christoffel table into Lecture 17's curvature formula and lowering the first index gives the independent nonzero components

$$(R_{0101}, R_{0202}, R_{0303}) = \frac{M}{r^3}(2, -1, -1),$$
$$(R_{1212}, R_{1313}, R_{2323}) = \frac{M}{r^3}(1, 1, -2).$$

All components not obtained from these by the curvature symmetries vanish. Raising all four indices leaves each listed component unchanged, because each index occurs twice. The two pair antisymmetries give four equal squares for each listed component. Thus

$$R_{abcd}R^{abcd} = 4(4 + 1 + 1 + 1 + 1 + 4)\frac{M^2}{r^6} = \frac{48M^2}{r^6}.$$

The ingoing metric and this scalar are analytic for $r > 0$, so the identity extends across $r = 2M$. The scalar diverges as $r \to 0$. This curvature singularity cannot be removed by a change of coordinates.

19.5 The massive orbit equation

Consider a future timelike Schwarzschild geodesic parametrized by proper time. Rotations of the angular coordinates are isometries because they preserve the round-sphere factor. Rotate them so the initial position and tangential velocity lie in the equatorial plane. The θ equation has the solution $\theta = \pi/2$, $\dot\theta = 0$ for these initial data. Uniqueness of the geodesic equation keeps the orbit in that plane. The constants E and L are its energy and angular momentum per unit rest mass.

Proposition 19.11 (Schwarzschild orbits · *proved*)**.** *A proper-time-parametrized equatorial timelike geodesic obeys*

$$\dot r^2 + f\left(1 + \frac{L^2}{r^2}\right) = E^2.$$

If $L \neq 0$, use ϕ as parameter and put $u = 1/r$. Then, with primes now denoting derivatives with respect to ϕ,

$$u'' + u = \frac{M}{L^2} + 3Mu^2.$$

Proof. The normalization $g(\dot\gamma, \dot\gamma) = 1$ gives

$$1 = f\dot t^2 - f^{-1}\dot r^2 - r^2\dot\phi^2 = \frac{E^2 - \dot r^2}{f} - \frac{L^2}{r^2}.$$

Rearranging gives the radial energy equation. To retain turning points, use the radial geodesic equation itself:

$$\ddot r = -\frac{ff'}{2}\dot t^2 + \frac{f'}{2f}\dot r^2 + rf\dot\phi^2 = -\frac{M}{r^2} + \frac{L^2}{r^3} - \frac{3ML^2}{r^4}.$$

The last equality substitutes the constants and the normalization just proved. It never divides by $\dot r$. Since $\dot\phi = Lu^2 \neq 0$, the chain rule gives $\dot r = -Lu'$ and $\ddot r = -L^2u^2u''$. Substitute in the radial equation and divide by $-L^2u^2$ to obtain the result. ∎

The Newtonian orbit equation lacks the term $3Mu^2$. We can see its effect perturbatively. Put $p = L^2/M$, $\epsilon_{\mathrm{orb}} = M/p$ and $w = pu$. The orbit equation becomes

$$w'' + w = 1 + 3\epsilon_{\mathrm{orb}} w^2.$$

For $0 < e < 1$ and $\epsilon_{\mathrm{orb}} \ll 1$, the Newtonian ellipse is $w_0 = 1 + e\cos\phi$. Writing $w = w_0 + \epsilon_{\mathrm{orb}} w_1 + O(\epsilon_{\mathrm{orb}}^2)$ gives

$$w_1'' + w_1 = 3w_0^2 = 3 + \frac{3}{2}e^2 + 6e\cos\phi + \frac{3}{2}e^2\cos(2\phi).$$

One particular solution is

$$w_1 = 3 + \frac{3}{2}e^2 - \frac{1}{2}e^2\cos(2\phi) + 3e\phi\sin\phi.$$

The term growing with ϕ is the first-order expansion of a shifted phase:

$$\cos((1 - 3\epsilon_{\mathrm{orb}})\phi) = \cos\phi + 3\epsilon_{\mathrm{orb}}\phi\sin\phi + O(\epsilon_{\mathrm{orb}}^2).$$

The remaining displayed terms in w_1 are bounded. Thus successive closest approaches, called perihelia, are separated by $2\pi/(1 - 3\epsilon_{\mathrm{orb}})$ to first order. The advance beyond a full turn is

$$\Delta\phi = 6\pi\epsilon_{\mathrm{orb}} + O(\epsilon_{\mathrm{orb}}^2) = \frac{6\pi M^2}{L^2} + O(\epsilon_{\mathrm{orb}}^2).$$

This is a weak-field prediction for bound test-particle orbits. The scale $M = G_N m/c^2$ is a length, so the approximation requires $p \gg M$. The figure plots the phase approximation with its bounded first-order corrections.

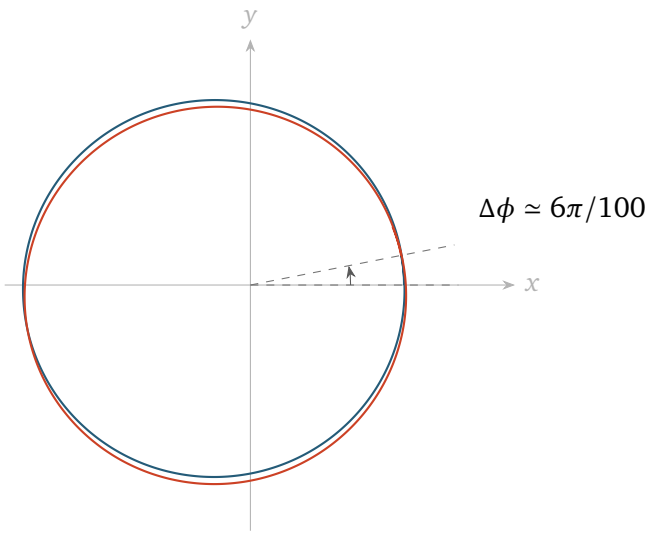


Figure 19.6: For $M = 1$, $p = 100$ and $e = 1/5$, the plotted first-order model is $w = 1 + e\cos((1 - 3\epsilon_{\mathrm{orb}})\phi) + \epsilon_{\mathrm{orb}}(3 + 3e^2/2 - e^2\cos(2\phi)/2)$, with $r = p/w$. The dashed rays mark successive perihelion directions to first order.

For radial motion $L = 0$, the radial energy equation still applies, but ϕ cannot serve as parameter.

19.6 A circular orbit and its effective potential

Definition 19.12 (Effective potential)**.** The *effective potential* for fixed L is

$$U_{\text{eff}}(r) = f(r)\left(1 + \frac{L^2}{r^2}\right).$$

The radial equation is $\dot{r}^2 + U_{\text{eff}} = E^2$ and $\ddot{r} = -U'_{\text{eff}}/2$. Radial motion is allowed exactly where $U_{\text{eff}} \le E^2$. For $M = 1$, $L = 4$ and $E^2 = 47/50$, the turning radii are the three roots $r_1 < r_2 < r_3$ of

$$3r^3 - 100r^2 + 800r - 1600 = 0.$$

They satisfy $2 < r_1 < 4 < r_2 < 12 < r_3 < 30$. The allowed exterior intervals are $2 < r \le r_1$ and $r_2 \le r \le r_3$. The outer interval describes a bound radial oscillation between closest and farthest approach. It does not imply a closed spatial orbit. For comparison, $E^2 = 51/50$ lies above the potential everywhere outside the horizon, since $1 - U_{\text{eff}} = 2(r-4)^2/r^3 \ge 0$ for these parameters. An inward orbit at this energy has no radial turning point before the horizon. The figure marks the energy levels and allowed intervals.

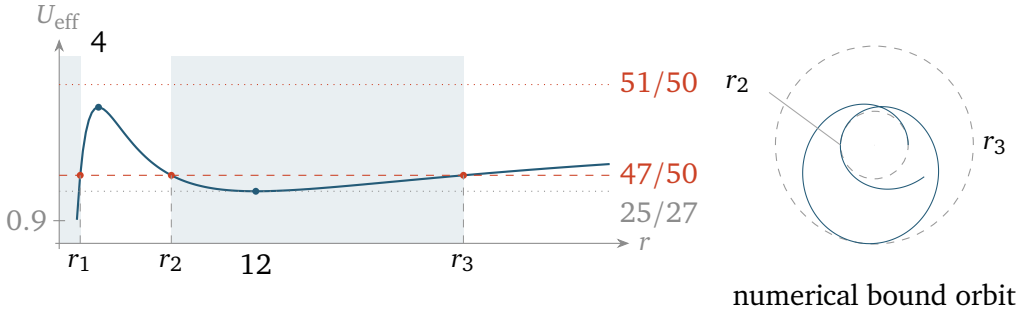


Figure 19.7: For $M = 1$ and $L = 4$, the potential has a maximum at $r = 4$ and a minimum at $r = 12$, with stable circular level $25/27$. The level $E^2 = 47/50$ gives three turning points and the shaded allowed radial intervals on the truncated potential plot, beside a numerical outer orbit between its turning circles.

Example 19.13 (A stable circular orbit)**.** Find a massive circular orbit at radius $r_0 = 12M$. For a general circular radius $r_0 > 3M$, solving $U'_{\text{eff}}(r_0) = 0$ and $E^2 = U_{\text{eff}}(r_0)$ gives

$$L^2 = \frac{Mr_0^2}{r_0 - 3M}, \qquad E^2 = \frac{(r_0 - 2M)^2}{r_0(r_0 - 3M)}.$$

With the angular momentum held fixed, the second derivative there is

$$U''_{\text{eff}}(r_0) = \frac{2M(r_0 - 6M)}{r_0^3(r_0 - 3M)}.$$

It is positive for $r_0 > 6M$, so the radial linearization has a restoring force. At $r_0 = 12M$ the constants are $L = 4M$ and $E^2 = 25/27$. This gives the potential minimum marked in the figure after setting $M = 1$.

Self-check. Recover the inward Newtonian acceleration from Γ^r_{tt} far from the Schwarzschild source.

Answer. The table gives $\Gamma^r_{tt} = ff'/2 = M(r-2M)/r^3 = M/r^2 + O(M^2/r^3)$. For slow motion, the geodesic equation contributes $-\Gamma^r_{tt}$ to the coordinate-time radial acceleration. Its leading value is $-M/r^2$, the attractive Newtonian result. It also matches $-\partial_r\Phi$ for $\Phi = -M/r$.

Self-check. A proper-time-parametrized equatorial timelike geodesic has $L = 4M$ and a radial turning point at $r = 9M$. Find E^2, $\ddot{r}$ and u'' there. Determine whether the turning point is a closest or farthest approach.

Answer. At the turning point $\dot{r} = 0$, so normalization gives $E^2 = (1-2/9)(1+16/81) = 679/729$. We use the radial geodesic equation, since differentiating the energy equation would give $0 = 0$ at this point:

$$\ddot{r} = -\frac{1}{81M} + \frac{16}{729M} - \frac{48}{6561M} = \frac{5}{2187M} > 0.$$

The chain rule $\ddot{r} = -L^2u^2u''$ then gives $u'' = -5/(432M)$. The positive radial acceleration makes this a closest approach.

Summary. A weak static Lorentzian metric gives Newton's equation for slow free fall. The cited Einstein–Hilbert variation gives Einstein's equation with positive $8\pi G_N T_{\mu\nu}$ in our conventions. Contracting the second Bianchi identity gives covariant conservation. The Schwarzschild exterior has zero Ricci tensor and nonzero full curvature. Its Killing constants, unit timelike normalization and radial geodesic equation give the massive orbit equation $u'' + u = M/L^2 + 3Mu^2$ in geometric units, for $L \neq 0$. Lecture 20 describes fields that require several local frames.

Reading. Nakahara, §7.10.1–2 and Example 7.15. Carroll, *Lecture Notes on General Relativity* (1997), §4, especially pp. 104–116, and §7, pp. 172–185.